5(a)(i)	infrared	**B1**
5(a)(ii)	$v = f\lambda$ **or** $c = f\lambda$	**C1**
	$f = 3.0 \times 10^{8} / 8.4 \times 10^{-6}$ $= 3.6 \times 10^{13}$ (Hz) $= 36$ THz	**A1**
5(b)(i)	constant phase difference (between the waves) (with time)	**B1**
5(b)(ii)	$\lambda = ax / D$	**C1**
	$x = 22 / 8$ **or** 2.75 (mm) **or** $22 \times 10^{-3} / 8$ **or** 2.75×10^{-3} (m)	**C1**
	$a = (6.2 \times 10^{-7} \times 2.8) / (22 \times 10^{-3} / 8)$ $= 6.3 \times 10^{-4}$ m	**A1**
5(c)(i)	difference in distances $= 6.2 \times 10^{-7} / 2$ $= 3.1 \times 10^{-7}$ m	**A1**
5(c)(ii)	phase difference = 360°	**A1**